



RMIT UNIVERSITY

Subject Code	ECON11558
Subject Name	Frontiers of the Digital Economy
Campus	SGS Campus
Title of Assignment	Whitepaper Report – Critical Economic Analysis of Blockchain Application
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Assignment Due Date	16/01/2026
Date of Submission	16/01/2026
Number of Pages	10
Word Count	1309

I declare that in submitting all work for this assessment, I have read, understood, and agree to the content and expectations of the Assessment Declaration.

ROYAL MELBOURNE INSTITUTE OF TECHNOLOGY

SGS CAMPUS

ECON1558 – Frontiers of the Digital Economy

Assignment 3

WHITEPAPER REPORT – CRITICAL ECONOMIC ANALYSIS OF BLOCKCHAIN APPLICATION

Case Study: Power Ledger

Word Count: 1309

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Sem C, 16 January 2026

Instructor: Dr. Mai Bui

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Abbreviation

CfD	Contract-for-difference
DER	Distributed Energy Resources
DPPA	Direct Power Purchase Agreement
DSO	Distribution System Operators
EVNCPC	Viet Nam Electricity Central Power Corporation
FIT	Feed-in-tariff
P2P	Peer-to-peer
PoS	Proof-of-Stake

I. INDUSTRY OVERVIEW AND PROBLEMS ARISE

1. INDUSTRY OVERVIEW

Distributed Energy Resources (DERs), including rooftop PV, batteries, and microgrids, enable consumers to become active prosumers, shifting electricity systems toward bidirectional energy flows (Zedan et al. 2024; Karandikar et al. 2021; Power Ledger 2019). Yet, the current retail market arrangements still largely operate on centralized energy generation and passive consumption, introducing friction and inequities in traditional market design.

2. INCENTIVE MISALIGNMENT AND RESOURCES UNDERUTILIZATION

The first issue is that **renewable energy is being stranded, weakening prosumers' incentives** under capped offtake. With self-consumption below 30%, a significant remaining share would often be exported, **aiding grid parity** (Gautier et al. 2018). In retail markets, surplus energy is purchased solely by DSOs under administrative agreements (Guarasico 2025). For example, under Vietnamese Decree 135/2024/ND-CP, excess sales to the national grid are capped at 20% of installed capacity and priced at the average retail tariff (Tan 2024). This means surplus exceeding the cap is curtailed and wasted, which under-utilizes renewable energy and further limits the return on DER investments.

3. HIGH TRANSACTION COSTS

Contract agreements between the prosumers and DSOs discussed above present a “principal-agent” setting, raising two additional asymmetric information problems. Ex-ante, **hidden characteristics**, where limited prosumers' information existed, DSOs risk selecting adverse exporters. Ex-post, **hidden action** and **hidden information**, introduce moral hazard risks, as prosumers undertake strategic behind-the-meter actions to maximize payouts (Wasa et al. 2019). These opportunistic behaviors further create an **opaque environment** and **trust deficit**. To mitigate these information gaps, more granular settlement data (30-minute to 5-minute) is required (Mwampashi et al. 2024). However, the increased frequency of intervals (48 to 288 per day) raises overall governance transaction costs associated with monitoring and reconciling (AEMO 2020).

4. POWER LEDGER INTRODUCTION

Power Ledger proposes a transparent energy-trading platform for a market-driven grid that expands consumers' choice and supports ‘democratization of power’ (Power Ledger 2019). This platform

leverages blockchain, a “trust machine”, to facilitate financial settlement of transactions between Application Hosts (e.g., microgrid operators, retailers) and end Participants, including prosumers and consumers, within the same trading intervals in which energy is produced and consumed (Power Ledger 2019).

II. BLOCKCHAIN PROJECT’S DESIGN ADDRESS ECONOMIC PROBLEM

To facilitate trading, the platform employs a dual-token design, with POWR serving as software-licensing tokens and Sparkz as fiat-pegged local settlement tokens (Power Ledger 2019). Moreover, Sparkz issuance is supported by an Ethereum SmartBond escrow structure used by Application Hosts.

For Power Ledger’s operational layers, the stack combines an Ethereum public layer that anchors the Power Ledger core with an EcoChain consortium chain and state channels to achieve high throughput (Power Ledger 2019). Finally, FuseBox, a configurable P2P application, was designed to fit the regulated market structure and ensure compliance with regulatory changes (Power Ledger 2019).

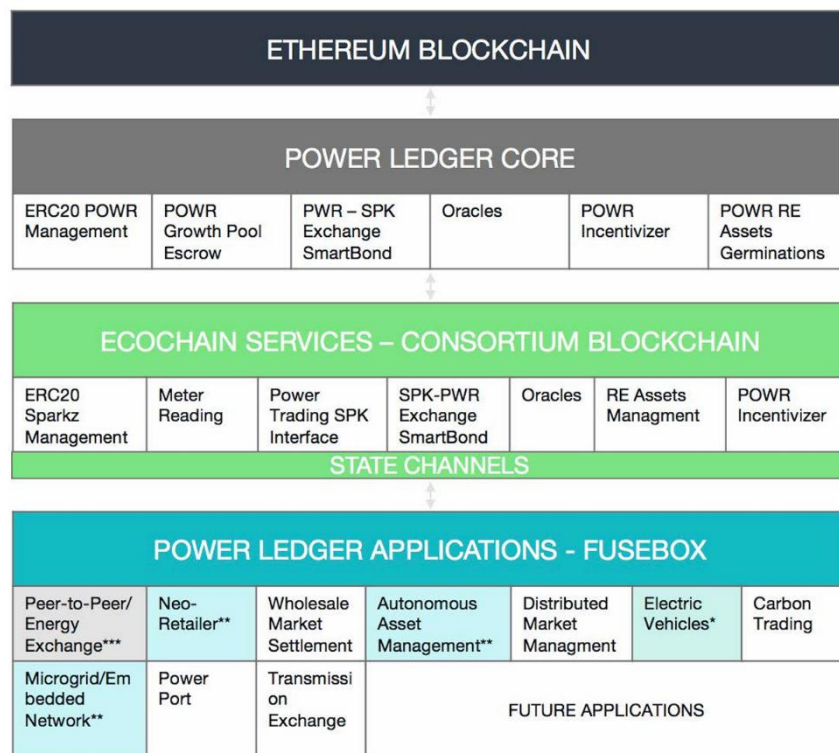


Figure 1: Power Ledger’s design (Source: Power Ledger 2019)

These designs make Power Ledger serve as a layer market institution that specifies access control, data validation, trade matching, and settlement decisions. All of this is supported by five key blockchain features: decentralization, distribution, consensus, immutability, and programmability. These features power the platform applications that further help to address moral-hazard risks, frictions, and inequities.

Regarding the first problem, the binding constraint is regulatory; therefore, Power Ledger cannot override the capped export volume or the buyback structure. Instead, the platform reduces renewable waste and mitigates disincentives by shifting from **mandatory grid exports to local matching**. Power Ledger's P2P trading application (FuseBox) uses a **trade engine** that continuously matches local surplus supply and demand, filling orders in equal increments and prioritizing proximity groups (Power Ledger 2019). With this institutional design, the platform offers higher utilization of surplus, reducing curtailment, increasing and diversifying prosumers' revenue streams, and improving perceived fairness, all without changing the regulated export cap.

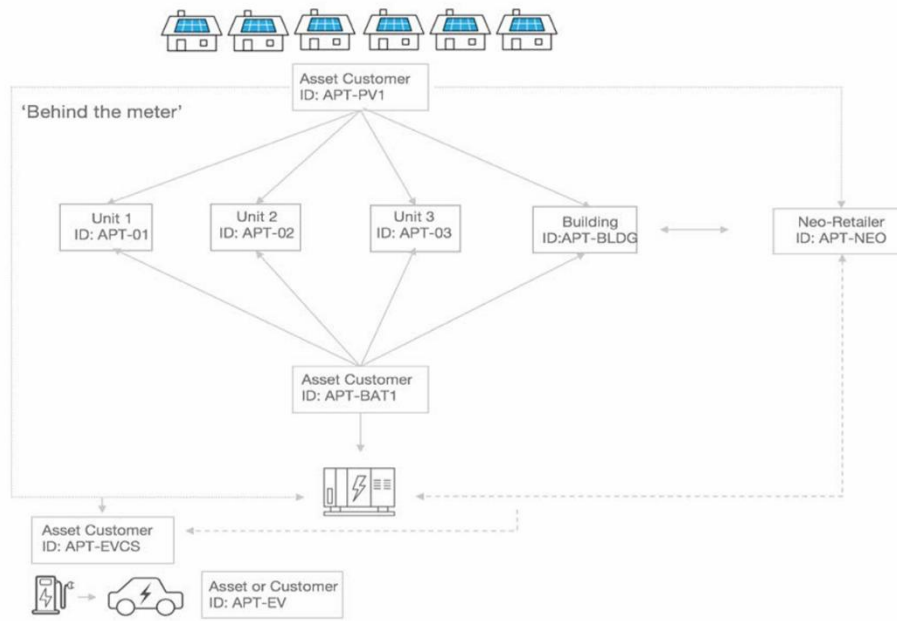


Figure 2: FuseBox's trade engine mechanism (Source: Power Ledger 2019)

Moreover, the **Green Energy Loyalty Reward Program**, under which a small fee on P2P transactions funds POWR purchases, which are then redistributed, and **Consumer Choices**, a voluntary green premium, are all enabled by the **programmability** feature of blockchain, in which rule-bound incentives are encoded into the platform (Power Ledger 2019).

For the second problem, Power Ledger can mitigate moral-hazard risks arising from information asymmetry by making contract performance more verifiable. EcoChain acts as a **permissioned shared ledger** with **programmability**. Metering readings recorded at configurable intervals (1-30 minutes) by Fusebox are written directly to EcoChain, which allows only authorized stakeholders (DSOs/retailers/prosumers) to reference the same **immutable** audit trail. Energy transactions are now settled against a common record rather than via unilateral spreadsheets via POS **consensus** on the consortium layer (Power Ledger 2019). Moreover, to **support high-frequency settlement**, State

Channels update balances off-chain, while oracles post a mutually agreed final state on-chain, preserving auditability without excessive reconciliation overhead (Power Ledger 2019).

III. LIMITATION AND REAL-WORLD ADOPTION ASSESSMENT

1. FEATURE' S LIMITATION

Despite the benefits blockchain has brought, there remain limitations to its capacity to address the identified problems.

For the first problem, Power Ledger operates in accordance with the country's retail regulations. In Vietnam, as Decree 135/2024/ND-CP **caps surplus sales to the grid** and pegs payments to the prior-year average market price, with **no time-specific value reflected**, Power Ledger cannot lift the cap or redesign export incentives (Tan 2024; Power Ledger 2019). In contrast, Australia has **time-of-use export tariffs**, and FueBox is configured to the local pricing structure to **pass through those incentives**. Moreover, Power Ledger's **geo-location priority feature is underdeveloped**, as mentioned in its white paper, leaving local matching applications not yet fully implemented in the platform's trading engine (Power Ledger 2019).

Regarding principal-agent frictions reduction, although immutability is provided, blockchain technology **does not guarantee input integrity**. Therefore, reliable oracles (smart meters) are required to deliver accurate off-chain dense injections and curtailment events for off-chain settlement at state channels, and then imported on-chain with other reliable oracles. Moreover, blockchain's immutability can raise concerns regarding data protection laws; in this case, Power Ledger could violate the "**right to be forgotten**" under PDPL (Vietnam) and the GDPR (EU) (Almeida and Soest 2021).

2. ALTERNATIVE SOLUTIONS COMPARISON

Under Decree 80/2024/ND-CP, Vietnam's current institutional solution for DDPA is a stacked-contract settlement in which prosumers sign a PPA with EVN, EVN signs a PPA with the consumer, and the prosumers and the consumer sign a CfD to reconcile the agreed-upon price (Bellini 2024). This structure once again creates information asymmetry and reconciliation frictions, as meter readings can diverge across parties' ledgers. This makes blockchain a necessary technology as it provides a distributed ledger and immutable records, which provide auditable trails, further reducing disputes and monitoring costs.

3. REAL-WORLD DEPLOYMENT

Power Ledger has partnered with EVNCPC to launch Vietnam's first P2P energy trading project, beginning in March 2023 with a 6-month trial to test the platform's effectiveness in reducing energy costs, improving price transparency, and providing a foundation for future policy decisions (EVN 2023; Power Ledger n.d.). However, to date, there has been limited news regarding the project's continuity.

IV. REFLECTION ON GUEST LECTURE

In the traditional financial system, trust is established by intermediaries (banks, clearinghouses, custodians) to verify identities, ownership, and balances. In digital asset transactions, trust has become even more important because transactions are irreversible.

In the session, Mr. Roshan emphasized that blockchain, validators, and custodians can serve as a trust stake in a digital asset. Leveraging blockchain, its programmability enables trust in mathematics rather than intermediaries; in other words, trust shifts from institutions to code. Validators then strengthen this framework by independently verifying transactions with an enforcement mechanism to incentivize honest behavior, such as staking tokens to become validators or slashing for dishonest behavior.

Regarding custodians, I find it interesting when Mr. Roshan quotes: "If blockchain removes intermediaries, why do we need custodians?". So custodians complement on-chain trust by enabling institutional participation through private-key management, asset segregation, and regulatory compliance, enabling usable at scale. These three components create a trust ecosystem for trading digital assets.

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